

MedTrack — Dashboard Storyboard

Milestone 2 — Module 4: Dashboard Planning & Prototyping

Objective: Design an interactive Tableau dashboard structure that converts the Tableau-ready hospital dataset into clear, decision-oriented healthcare insights.

Data source: data/processed/hospital_final_dataset.xlsx (200 records, 26 columns)

Global Filters & Interactivity

Global element	Planned implementation
Date Range	Filter dashboard data by admission date
Department	Filter or compare hospital departments
Ward	Used where ward-level resource analysis is relevant
Treatment Type	Used for patient-flow analysis
Navigation	Overview as landing page with access to the other three dashboards

1. Hospital Overview

Purpose: Provide management with a high-level snapshot of hospital performance.

Section	Planned content
KPI cards	• Total Admissions • Occupancy Rate • Average Length of Stay (ALOS) • Readmission Rate • Bed Utilization Rate
Visualizations	• Admissions trend over time • Admissions by department • Department efficiency comparison
Filters	Date Range, Department
Layout	KPI cards at the top; primary trend/comparison visual in the central area; supporting visuals arranged below.

2. Patient Flow

Purpose: Understand patient movement, treatment patterns, length of stay and readmissions.

Section	Planned content
KPI cards	• Total Admissions • Average Length of Stay • Readmission Rate
Visualizations	• Admissions trend • Patients by ward • Treatment Type distribution • Surgery vs non-surgery • Readmission analysis
Filters	Date Range, Department, Ward, Treatment Type
Layout	KPI cards at the top; primary trend/comparison visual in the central area; supporting visuals arranged below.

3. Department Analytics

Purpose: Compare departments using workload, efficiency, stay duration, readmission and satisfaction measures.

Section	Planned content
KPI cards	• Department Admissions • Department Efficiency Score • Average Length of Stay • Readmission Rate • Patient Satisfaction
Visualizations	• Admissions by department • Average length of stay by department • Efficiency score by department • Readmission rate by department • Patient satisfaction by department
Filters	Department, Date Range
Layout	KPI cards at the top; primary trend/comparison visual in the central area; supporting visuals arranged below.

4. Resource Utilization

Purpose: Monitor bed usage and operational resource pressure across wards and departments.

Section	Planned content
KPI cards	• Bed Utilization Rate • Occupied Beds • Available Beds • Average Length of Stay
Visualizations	• Occupancy by ward • Wait time by department • Average length of stay • Surgery workload • Bed status distribution
Filters	Ward, Department, Date Range
Layout	KPI cards at the top; primary trend/comparison visual in the central area; supporting visuals arranged below.

Dashboard Navigation & Prototype Plan

The Hospital Overview dashboard is planned as the landing page. Navigation controls will provide access to Patient Flow, Department Analytics, and Resource Utilization. Filters should update the relevant views consistently.

Prototype acceptance checks

- All four planned dashboards are represented.
- Core KPI cards display values from the Tableau-ready dataset.
- Filters interact with the relevant views.
- Charts have clear titles and readable labels.
- Navigation between dashboard views works.
- The final Tableau workbook is saved as medtrack_prototype.twbx.